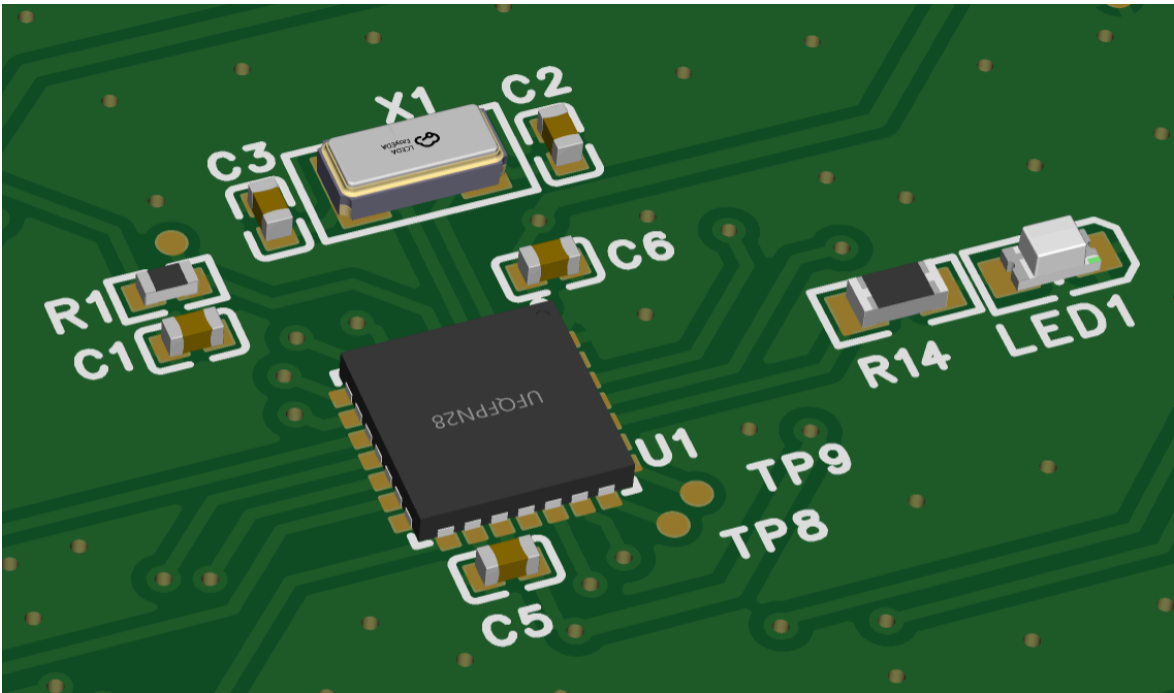
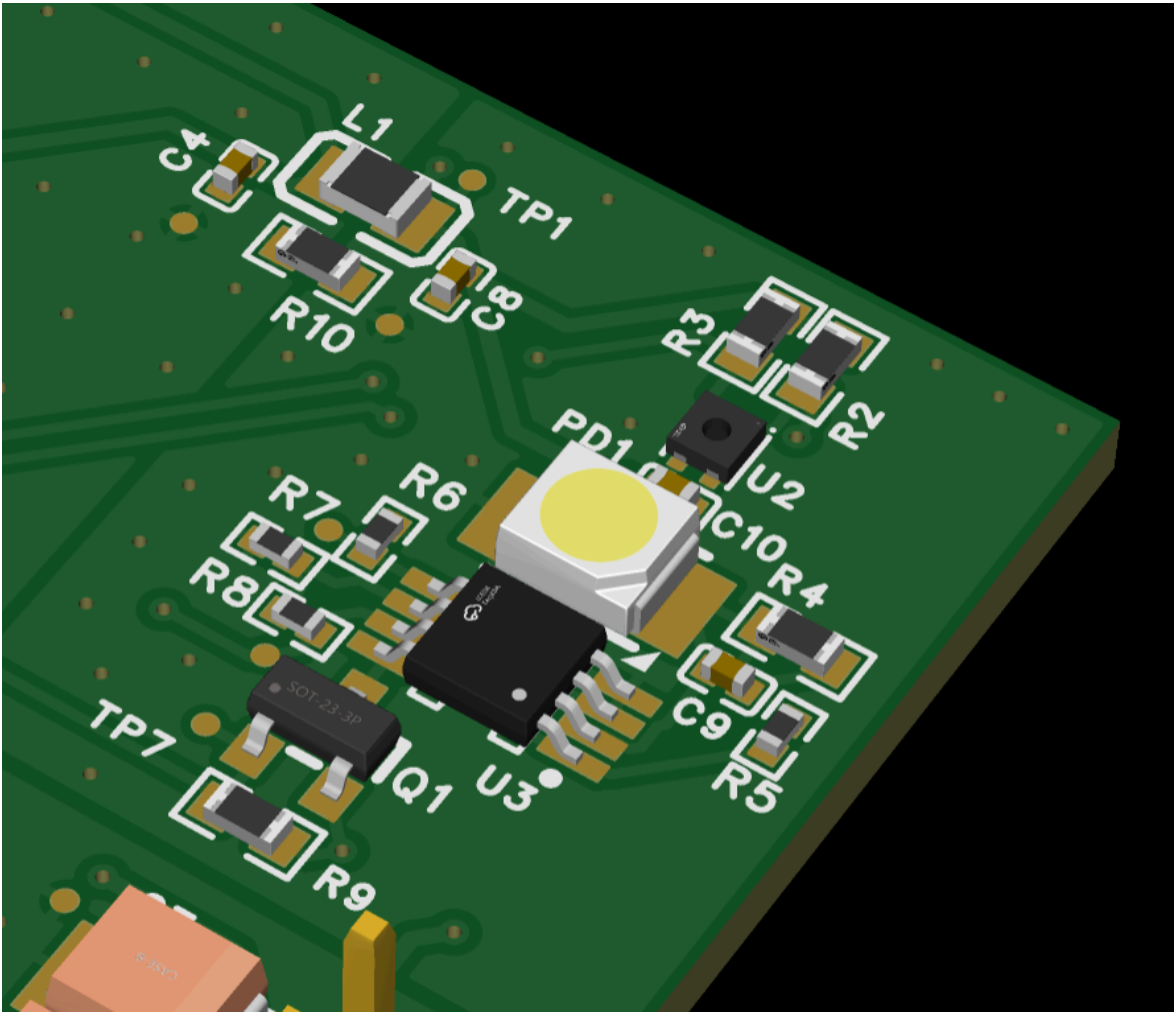


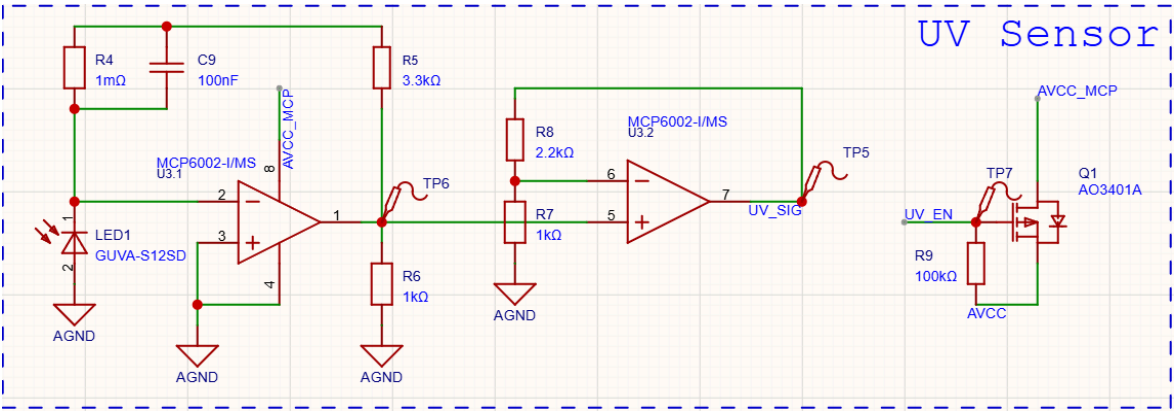
MCU: STM32L031G6U6

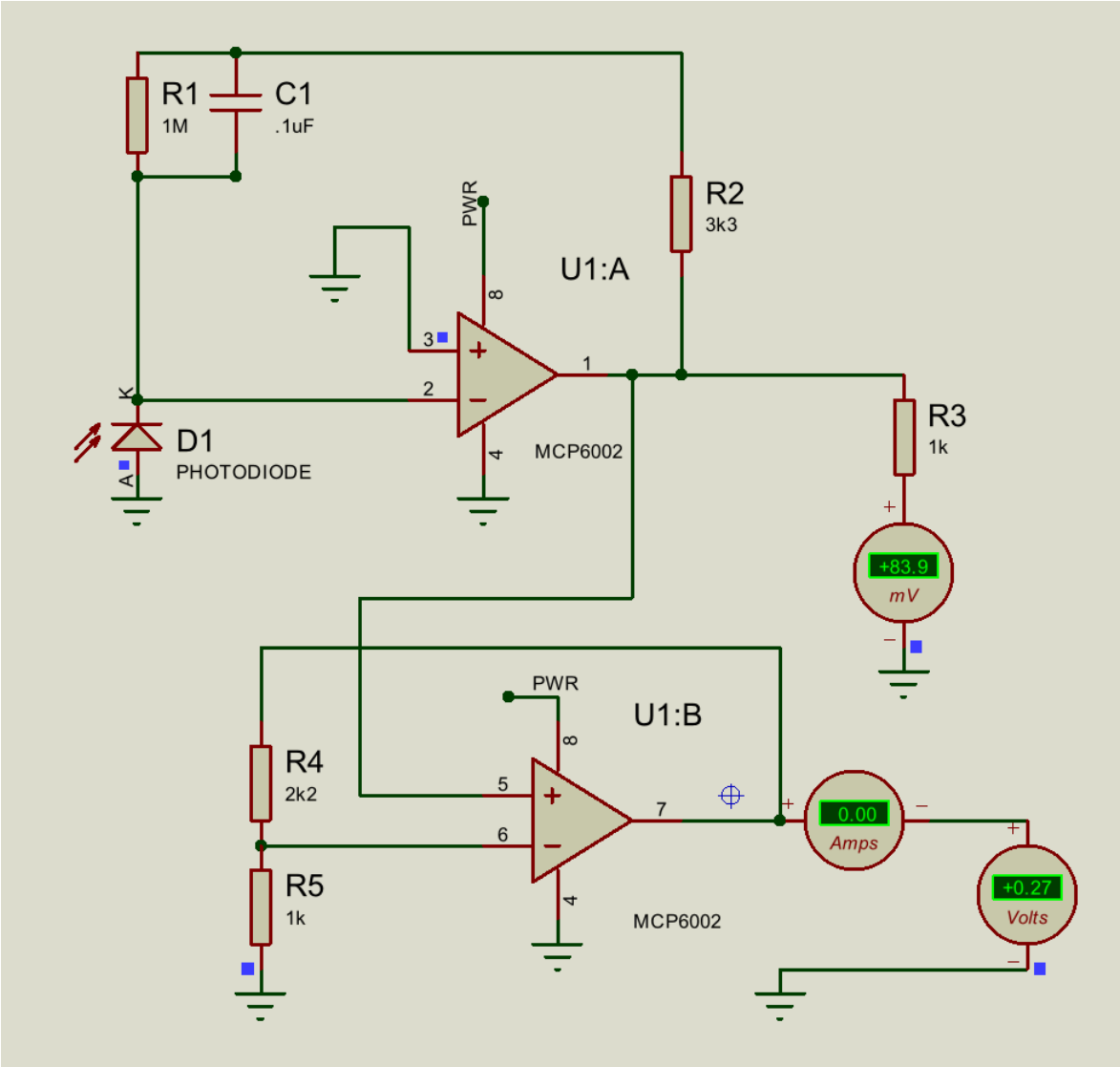


Sensors



UV SENSOR: GUYA-S12SD





GUVA-S12SD Simulation Data

Sensor (GUVA-S12SD UV Photodiode):

- Converts ultraviolet irradiance into a proportional photocurrent (≈ 21nA per UV Index unit).
- Designed for an expected operating range of 0nA to 340nA (covering standard UVI 0 through extreme outdoor conditions).

Stage 1 (U1A: Transimpedance Amplifier & Attenuator):

- **TIA Conversion:** Converts input photocurrent to voltage via the 1MΩ feedback resistor R1, paired with a nF capacitor C1 for high-frequency noise filtering.
- **Attenuation Network:** Uses a resistive voltage divider (R2 = 3.3kΩ, R3 = 1kΩ) to scale the intermediate voltage level prior to the second stage.

Stage 2 (U1B: Non-Inverting Buffer & Amplifier):

- Amplifies the scaled intermediate signal with a fixed voltage gain of 3.2:

$$Gain = 1 + \frac{R4}{R5} = 1 + \frac{2.200}{1.000} = 1 + 2.2 = 3.2$$

- Buffers the final analog output to prevent loading effects and maps the sensor range safely within the microcontroller ADC's operating window below the battery rail VDD.

ADC Stage & Conversion:

- Configured for the **STM32L031 microcontroller ADC**, sampling the final analog output voltage on an analog input channel.
- Employs **ratiometric software compensation** via the internal reference channel (VREFINT ≈ 1.22V) to completely eliminate measurement drift caused by coin cell voltage sag over time.

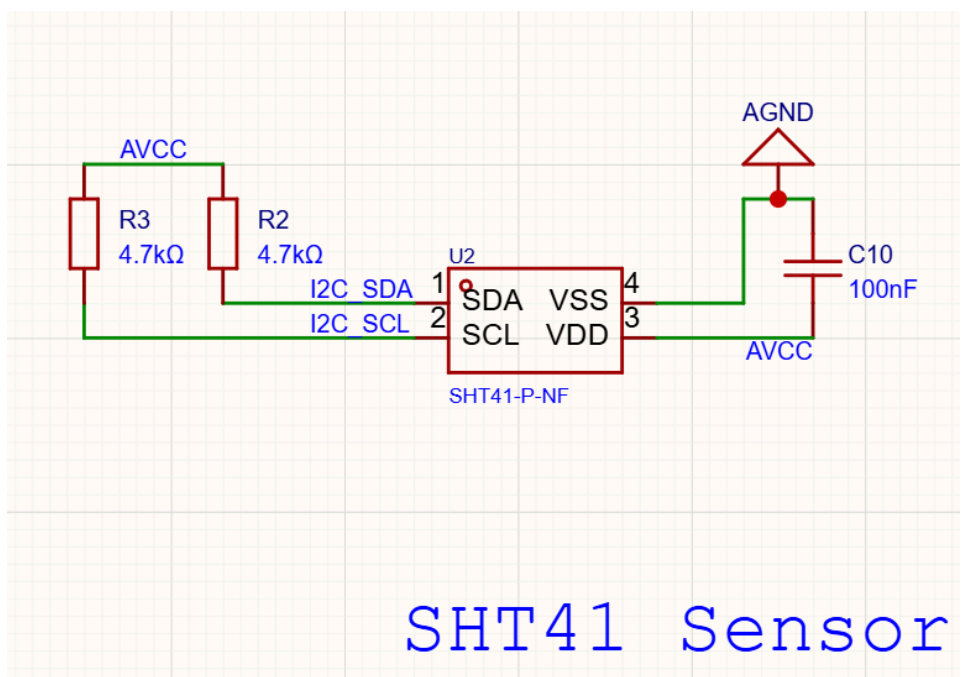
UV Index Calculation Formula:

- Converts raw digital ADC counts into a true, battery-independent UV Index value using the formula:

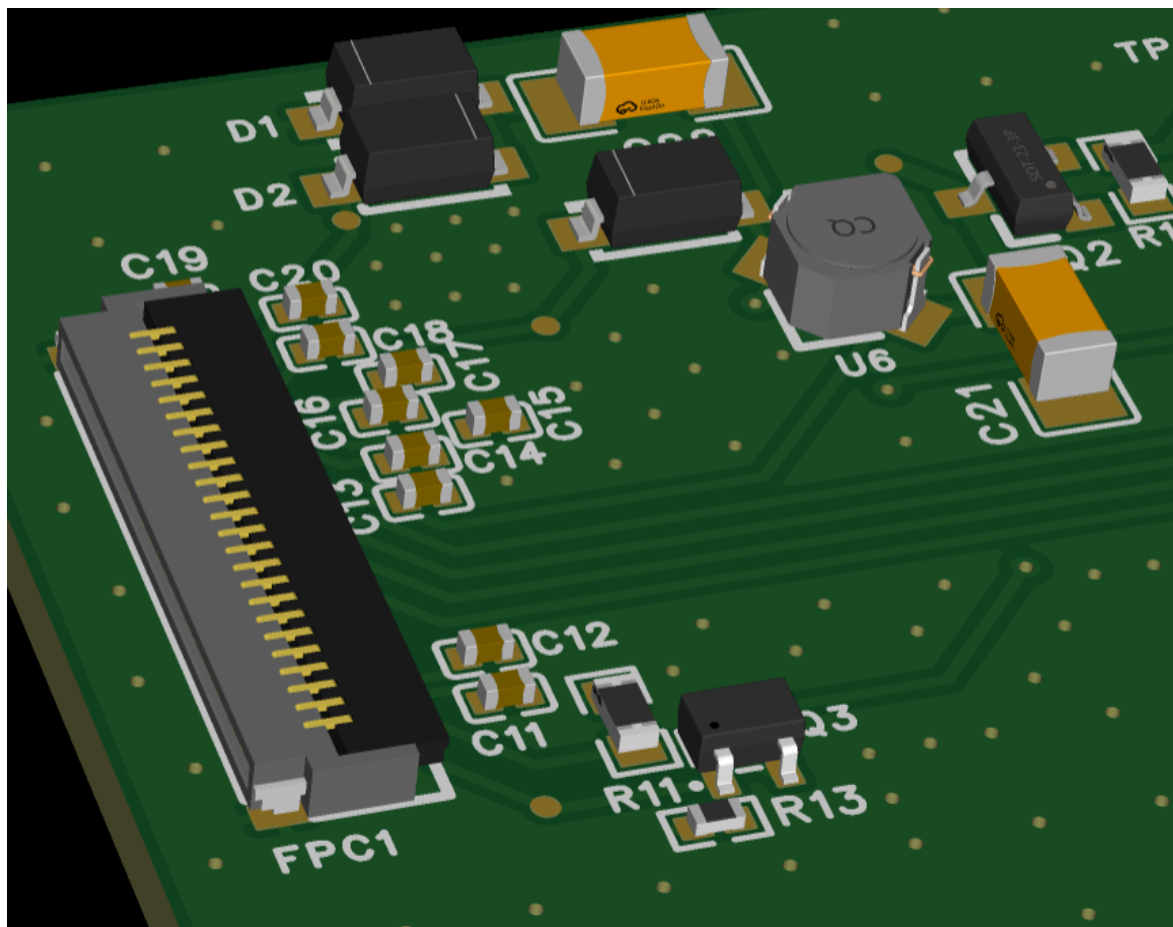
$$UVI = \frac{ADC_{RAW} \times V_{REFINT-CAL}}{4096 \times GAIN \times 21 \times 10^{-3}}$$

- Maps the normalized voltage directly to the sensor's linear sensitivity curve (≈ 21nA per UVI unit, scaled by the 1MΩ TIA and the 3.2 second-stage gain)

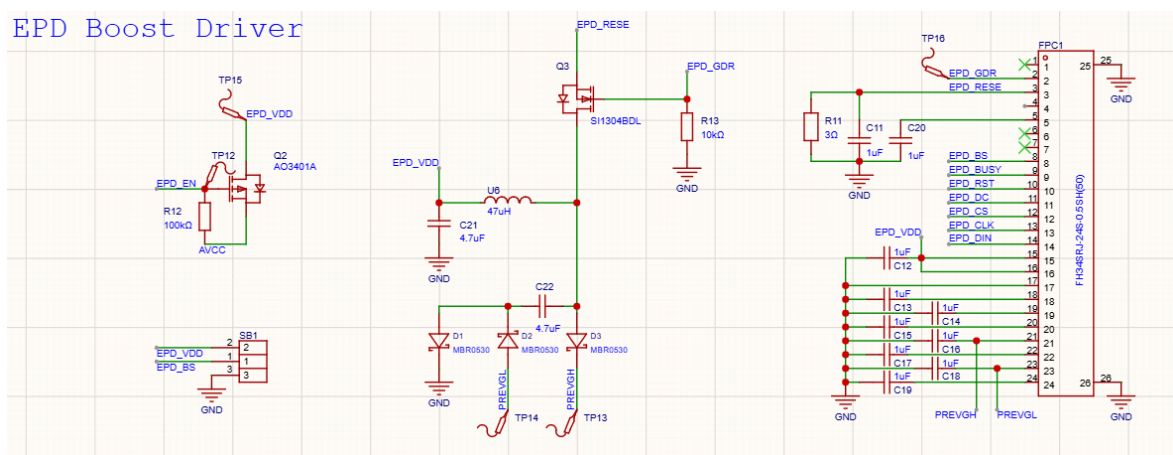
Temperature & Humidity: SHT-41



E-Ink Driver



EPD Boost Driver



[epd-datasheet/PINOUT/SPI-24PIN.md at main · CursedHardware/epd-datasheet](#)

USB Interface

